

- Process and Product Assurance, 334
- Process Areas
 - for the Integrated CMM, 239
 - grouping for Integrated CMM, 241
 - selection for CMM Integrated, 339
- Process Engineering Group (PEG), 76
- Process Focus, 327
- Process Improvement, 35, 80, 93, 200, 326, 339
 - stages of, 119
 - models and frameworks for, 94, 224, 319
 - planning for, 335
 - program, 337
 - proposals, 335
- Process orientation, symptoms & behaviour, 327, 328, 330
- Processes in ISO 12207, 246
- Processes-to-Business Objectives linkages, 335
- Processing attributes, 84
- Producer—in the review, 161
- Producer—responsibilities, 161
- Product
 - assurance, 318
 - attributes, 82
 - certification, 49
 - complaints, 318
 - metrics, 11, 114, 135
 - quality, 80, 82, 106, 126
 - registration procedures, 317
- Professional certifications, 321
- Project
 - de-briefing, 46
 - effort tracking, 136
 - management Issues in Maintenance Projects, 331
 - monitoring, 39
 - productivity, 38
 - S-curve, 138
 - tracking—use of metrics, 138
- Prototyping, 40, 368
- PSP and TSP for small organizations, 337
- Q**
 - QA strategies, 316
 - QAI (Quality Assurance Institute), 323
 - Certifications from, 323
 - CSTE program from, 308
 - QC group, 316
 - QC professionals, 316
 - QMS—organizational expectations, 59
 - QMS (See Quality Management System)
 - Quality—in the context of software development process, 92
 - Quality & reliability, 287
 - Quality Assurance (QA), 314
 - social commitment, 55
 - QA Auditor, 318
 - QA Compliance Associate, 317
 - QA Engineer, 316
 - QA Function (SQA), 19
 - QA group, 268
 - QA Institute (QAI), 83
 - QA Plan, 56
 - Quality Certification, 38, 311, 320
 - Indian statistics, 341
 - certified Indian organizations—alphabetical list, 343
 - Quality circles, 157
 - Quality consultant, 44
 - Quality Control (QC), 318
 - QC Analyst, 317
 - QC Manager, 318
 - QC Tools, 362
 - Quality, 1
 - certifications—from ASQ, 321, 322
 - criteria for, 86
 - culture, 47, 330
 - definition—ISO 9126, 87
 - engineer, 317
 - factors for, 85
 - gates for, 82
 - goal—DRE (Defect Removal Efficiency), 134
 - goal setting—sources of data, 133
 - IEEE definition, 88
 - improvement program—principles, 93
 - improvement program—principles, 93
 - improvement, 224
 - issues in Web-site maintenance, 380
 - management of—quantitative, 132, 202
 - metrics for, 11, 317
 - planning for—in the project, 40

- models for, 83
 - model—ISO 9126 Std, 87
 - objectives for, 56, 59
 - of support, 18
 - QFD (Quality Function Deployment), 36
 - QMS (Quality Management System), 25, 36, 37, 200, 214, 218
 - policy for, 25, 30
 - record for, 28
 - Quality Manual—contents, 25, 26
 - Quality maturity of the Indian software industry, 352
 - Quality systems, 27, 202
 - effectiveness of, 40
 - improvement in, 40
 - review of, 40
 - triangle of, 209
 - Quantitative quality management, 132
- R**
- Random cause, 362, 366
 - Random testing, 282
 - Recorder—responsibilities, 162
 - Regression testing, 295, 296
 - advantages of test automation, 296
 - Reliability, 83
 - testing for, 286
 - Representations for the CMM-I (Integrated CMM), 236, 237
 - Requirement, 9, 10, 385
 - bi-directional Traceability, 115, 387
 - creep in, 118
 - specification—the IEEE guide, 385
 - specification for, 37, 44
 - stability of (relationship to size), 118
 - stability index for, 116, 117
 - taxonomy for, 386
 - traceability, 43, 117
 - Re-use-Pack, 37, 43
 - Reviewer—responsibilities, 161
 - Reviews
 - ANSI Standard for, 60
 - compared with Audits, 60
 - compared with Inspections—roles and responsibilities, 159
 - cultural issues in, 165
 - educational qualities, 168
 - need for, 155
 - psychological aspects of, 162
 - team structure for, 168
 - types of, 260
 - Revisions v/s Versions, 181
 - Re-work, 16
 - Risk
 - identification, analysis and appraisal, 40, 41
 - management in High Management Organizations, 332
 - risk-based approach to testing, 292, 293, 294
 - Role of ISO consultants, 219
 - Role of mentor, 311
 - Role of Quality Assurance function, 20
 - Role of testing, 259
 - Ron Radice, 93
 - Root Cause Analysis (RCA)—concepts and template, 72, 73, 77
 - Round-robin reviews, 156
 - Rules for software configuration items selection, 178
- S**
- Schedule
 - as a measure, 136
 - performance index for (SPI), 138
 - slippage, 38
 - variance for (SV), 138
 - SCM (Software Configuration Management)
 - anti-patterns, 197
 - checklist for plan evaluation, 196
 - CMM aspects of, 193
 - IEEE definition for, 175, 177
 - major activities in, 177
 - people-ware, 197
 - pitfalls in, 197
 - S-curve, 138
 - Second-party-assessment and audit, 211, 212
 - Security features testing, 287, 317
 - Security technologies, 317
 - SEI (see Software Engineering Institute), 224
 - SEI
 - 1999 Survey (High Maturity Organizations), 331
 - CMM compliance, 320

- CMM Lead Assessor, 319
- CMM organization measures, 320
- Selecting ISO consultants, 218
- Self assessment, 211
- SEPG (Software Engineering Process Group)
 - Severity level, 63
- Shewhart-type control charts, 363
- Six Sigma, 329, 402
- Size
 - and requirements stability, 118
 - estimating in Maintenance Projects, 331, 332
 - metrics for, 11
- Skill level of project staff, 35
- Skills and Knowledge, 334
- Skills for SQA role, 50
- Small Organizations and CMM—
 - challenges, 335, 336
- Software application attributes—
 - classification, 84
- Software attributes, 82
- Software CMM
 - common features, 94
 - key process areas, 94
 - version 1.1 of, 225
 - for Small Organizations/projects, 335
- Software configuration audit, 188
- SCI (Software Configuration Item)
 - identification and rules for, 177, 179, 178
- SCM (Software Configuration Management), 183
 - need for, 173
 - description, 175
- Software Development Life Cycle (SDLC), 2
- Software Engineering Institute (SEI), 80, 93, 224
- Software life cycles, 214
- Software maintenance
 - definition, 110
 - metrics for, 110
 - models evolution, 367
- Software maturity, 224
- Software metrics, 11, 70, 96, 383
 - classification, 97
 - definition, 98
 - effectiveness attributes, 101
- Software practices maturity—Indian scenario, 341
- Software process
 - assessment, 210
 - assets, 67
 - capability, 211
 - defined, 67
 - engineering group for (SEPG), 47, 48
 - improvement (SPI), 211, 226, 319, 334
 - management of (SPM), 361
- Software product
 - characteristics of, 82
 - definition (IEEE Std. 610.12), 82
 - external attributes of, 83
 - internal attributes of, 83
 - model for quality—McCall, 86
 - quality, 83
 - quality—Boehm's view, 85
- Software project—umbrella activities, 102
- Software Quality
 - analyst (SQA), 47
 - assurance for (SQA), 333
 - assurance Manager, 319
 - assurance Plan—IEEE Standard, 56, 57
 - ISO 8402 definition, 88
 - metrics for, 88
 - tester for, 316
- Software requirements, 387
- Software standards—ISO, CMM, IEEE, 59
- Software systems evolution, 81
- Software testing
 - careers in, 258, 308
 - practices, 261
- Software verification, 318
- SPC (see Statistical Process Control)
- Special causes, 362, 366
- Specific practices in CMM-I, 242
- SPI (see Software Process Improvement), 226
- SQA (Software Quality Assurance)
 - SQA function, 19
 - benefits, 20
 - objectives, 22
 - people issues, 53
 - role and skills, 19, 20, 23, 50
 - need, 21

- SQA plans, 319
- SQA responsibilities, 23
- SRS (Software Requirements Specification)
 - formal notation, 18
 - IEEE Guide for, 61
- Stable process, 361
- Staged representation—Integrated CMM, 239, 240
- Stages v/s Continuous representation for CMM-I, 242
- Standard
 - process definition, 68
 - IEEE definition, 82
 - QAI definition, 83
 - procedures, 40
- State transition
 - analysis, 282
 - diagram, 62
- Statement coverage, 290
- Static testing, 283
- Statistical Process Control (SPC), 135, 360, 361
- Statistical Quality Control (SQC), 360, 361
- Statistics on Quality Certification scenario in India, 341
- Statistics on TickIT certifications, 245
- Status reports, 317
- Stress testing, 288, 315, 320
- Structure of the ISO 9001, 200
- Structured reviews, 168
- Structured walkthroughs, 156
- Sub-contractor, 65
- Suppliers—ISO 9001:2000 cause, 65
- Support quality, 18
- Surveillance audit, 215
- SW CMM
 - Ability to Perform, 225
 - Activities Performed, 225
 - Commitment to Perform, 225
 - Common Features, 225
 - comparison with CMM-I, 244
 - comparison with ISO, 227, 228
 - General Practices, 226
 - Key Practices, 225
 - Key Process Areas, 227, 234
 - mapping with ISO, 228
 - maturity levels in, 225
 - measurement & analysis in, 225
 - structure of, 226
 - Verifying Implementation, 225
 - mapping ISO/IEC 15504, 231, 232
 - v/s Integrated CMM, 235, 244
 - vis-à-vis ISO, 224
- SW CMM to CMM-I transition, 238, 239, 338
- System Integration tests, 315
- System Security Engineering CMM (SSE CMM), 252
- System tester, 314, 315
- System testing, 37, 41, 276
- T
 - Tailoring guidelines, 51, 68
 - Team debugging, 156
 - Team Software Process (TSP), 336
 - Technical maintenance—website, 377, 382
 - Technical reviews, 40, 168
 - Technology
 - change management (TCM), 43, 74
 - impact analysis, 74
 - innovation, 74
 - pilot—Summary Format, 75
 - Test analyst, 267, 268
 - Test artifacts, 269
 - Test automation, 297, 319
 - & Test Tool Selection, 266
 - during regression test—advantages, 296
 - Test bed, 271
 - Testing career—progression, 309
 - Test case, 37, 62, 271, 319
 - based on boundary value analysis, 281
 - effective, 260
 - execution, 268
 - generator, 271
 - prioritization, 267
 - specification, 271, 274
 - Test control metrics, 291
 - Test coverage & metrics, 271, 292
 - Test design specification, 274
 - Test documentation, 261, 270
 - Test driver, 271
 - Test factors, 294
 - Test harness, 271
 - Test life-cycle tools, 299
 - Test manager, 266
 - Test maturity, 258

- Test metrics, 289
 - Test observer, 268
 - Test phase matrix, 294
 - Test plan—261, 269, 273, 274, 294, 319
 - metrics, 292
 - preparation of, 264
 - template for, 274
 - Test procedure specification, 274
 - Test process evaluation & improvement, 265
 - Test Process Improvement (TPI), 258, 301
 - Test report, 271
 - Test resource metrics, 292
 - Test result metrics, 292
 - Test result record—format, 272
 - Test scripts and format, 267, 268, 272
 - Test specification document, 267, 269
 - Test strategies, 293, 294, 318
 - Test strategy development—steps, 294
 - Test summary report, 268
 - Test team leader, 267
 - Test tool selection and acquisition, 297
 - Testability, 84
 - metrics for, 291
 - Testing, 270, 311
 - additional roles in, 268, 269
 - competence for, 308
 - documentation & help, 285
 - human issues and challenges, 306
 - IEEE definition, 259
 - life cycle—264
 - maturity model (TMM), 301
 - policy for, 257
 - primary and secondary role, 307
 - product security, 317
 - professional certifications, 308
 - related definitions—IEEE, 270, 271
 - risk-based approach to, 292
 - role of, 259
 - security features, 317
 - strategies for, 257, 316
 - usability, 289
 - v/s Debugging, 262
 - v/s Inspection, 260
 - The need for ISO, 207
 - The Structure of ISO 9001, 201
 - The working of ISO 9001, 205
 - Third-party-audit and assessment, 211, 212
 - Thread testing, 283
 - TickIT
 - objectives, 245
 - purpose, 245
 - statistics on certification, 245
 - Tips for Author/Producer, 163
 - Tips for Reviewer, 163
 - TMM (Testing Maturity Model), 301
 - and CMM connection, 302
 - & automated software testing, 304
 - structure of, 303
 - Tom Gilb—on Walkthroughs, 158
 - Top twenty software companies in India, 349
 - Total quality culture (TQM), 54, 200, 316, 329
 - Traceability, 17, 62, 181
 - Traditional metrics—adaptation to OO paradigm, 152
 - Training, 42
 - responsibility of, 35
 - role in quality, 27
 - and awareness programs, 334
 - records of, 42
 - Transaction Log, 186
 - Transformability, 16
 - Transitioning from SW CMM to the CMM-I, 238, 239
 - Type I and Type II error, 363
 - Types of audits and assessments, 211
 - Types of CMMs, 234
 - Types of control charts, 365
 - Types of reviews, 260
- U**
- Unit test records, 40
 - Unit testing, 40, 275
 - Upgrading from SW CMM to CMM-I, 338
 - Upper and Lower Control Limits, 59
 - Usability, 83
 - User interface testing, 285
 - User satisfaction measures, 105
 - Using consultants, 217
 - Using the CMM—challenges for small organizations, 336
- V**
- Validation—IEEE definition, 258
 - Variable control charts, 365
 - Verifiability, 16

Index

Verification & validation, 9, 258
Version control systems, 315
Visibility, 181
Vision Statement, 25
V-Model for testing—benefits, 275
Volume testing, 288

W

Walkthroughs, 155
 and organization structure, 158
 as static QA methods, 157
 definitions, 157
 Myers' views, 157

 objectives, 157
 optimum number of participants,
 158
Walter Shewart, 93
Watts Humphrey, 93
WBS (see Work Breakdown Structure)
Web-site life cycle, 375, 376
Web-site maintenance and quality is-
 sues, 373, 380
Weighted methods per class, 149
White box testing, 273, 293, 295
Work breakdown structure (WBS), 138
Work product, 178
Working of ISO 9001, 200